

COMP 4421 (Spring 2025)

Assignment 1

Department of Computer Science and Engineering
The Hong Kong University of Science and Technology

Total = 100 marks

Due: 11:55 pm, March 26, 2026

Assignments must be submitted via CANVAS

Late Policy: Late assignments incur a 10%-score penalty; only one day late is allowed.

Programming Assignment

In this assignment, you should complete the Python functions in the given Jupyter Notebook. Download `programming1.ipynb` and related files from CANVAS.

1. Your submitted Jupyter Notebook must be named `programming1.ipynb`.
2. Runtime requirement: your notebook must finish within 5 minutes.
3. Compress all files into one file named
[your 8-digit student ID]_assign1.rar (or zip)
Example: 09654321_assign1.rar
4. If multiple submissions are made before the deadline, the latest version will be graded.
5. Plagiarism is taken seriously. You may discuss ideas but **must not share code or pseudocode**.
6. Marks will be deducted for requirement violations.

Grading criteria

- Correctness and efficiency of algorithms.
- Adherence to assignment. 20% penalty for any forbidden extra library imports.

Written Assignment

Submit your written assignment as a PDF named

[your 8-digit student ID]_w-assign1.pdf

Example: 09654321_w-assign1.pdf

Include this file in the same compressed submission.

Question W1: Histogram Equalization

Given an image I of size 5×5 pixels with intensity levels ranging from 0 to 7 (8 grey levels). Round your answers to **2 decimal places**, unless otherwise specified.

Input image

6	7	6	5	6
3	0	3	6	7
4	1	2	5	7
0	3	4	4	5
5	4	5	3	6

(a) Sketch the histogram for the image I .

(1) Perform histogram equalization based on the lecture notes procedure. Complete the intermediate table and round the final mapped intensity to the nearest integer.

Intensity Level k	0	1	2	3	4	5	6	7
Pixel Count n_k								
Probability $p(r_k) = n_k/25$								
Cumulative Probability $\sum_{j=0}^k p(r_j)$								
Mapping $s_k = 7 \sum_{j=0}^k p(r_j)$								
Final Intensity (Rounded s_k)								

Output Image

- (c) Histogram specification adjusts the histogram to match a specified probability distribution. Given the following distribution, perform histogram specification and fill in the output image in integer intensities.

Specified Distribution CDF

Target Intensity Level j	0	1	2	3	4	5	6	7
Target Probability $P_z(z_j)$	0.25	0.2	0.1	0.1	0.1	0.1	0.1	0.05
Target Cumulative $\sum_{i=0}^j P_z(z_i)$								

Intensity Mapping

Input Intensity r_k	0	1	2	3	4	5	6	7
Equalized Value $T(r_k)$								
Matched Output Intensity z_j								

Output Image

Question W2: Sobel Filtering

Given the following 7×7 grayscale image I . Use the corresponding 3×3 Sobel operators. Round your answers to **2 decimal places**.

0	0	0	0	0	0	0
1	0	0	0	0	0	0
1	1	0	0	0	0	0
1	1	1	<u>1</u>	0	0	0
1	1	1	1	1	0	0
1	1	1	1	1	1	0
1	1	1	1	1	1	1

- (a) Consider the 3×3 neighborhood centered at the underlined pixel 1. Compute the Sobel **horizontal gradients** G_x for each pixel in this 3×3 region.

- (b) Compute the Sobel **vertical gradients** G_y for the same 3×3 region.

- (c) Compute the **gradient magnitudes** $G = \sqrt{G_x^2 + G_y^2}$ for the same 3×3 region.

Question W3: Bilateral Filtering

Perform bilateral filtering on the input image. You do not need to fill in * cells because they correspond to zero-padding.

- Spatial sigma = 1
- Range sigma = 5
- Kernel size = 3

Round values to **2 decimal places**.

Input image

0	0	0	0	0	0
0	1	1	0	1	0
0	1	1	0	1	0
0	1	1	0	1	0
0	0	0	0	0	0

Output image

*	*	*	*	*	*
*					*
*					*
*					*
*	*	*	*	*	*